HHCKLA Buddhist Leung Chik Wai College

6A error analysis (2024)

Student No.: 06 Name: _____ Marks: / 55

Class: 6A Date: _____ Marked date: _____

What have you learned in this chapter? _____

What can you do to improve your work next time? _____

1. It is given that $g(x)$ is partly constant and partly varies as x. Suppose that $g(-3) = -21$ and $g(7) = 9$.

   (a) Find $g(x)$. (3 marks)

   (b) Let $h(x) = xg(x) + k$, where k is a real constant. If all the roots of the equation $h(x) = 0$ are real numbers, find the range of values of k. (3 marks)

   (6 marks) (DSE-2024 PI Q10)

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2. It is given that $\log_9 y$ is a linear function of $\log_3 x$. Denote the graph of the linear function by L. The slope of L is 4 and L passes through the point $(5, 22)$. Express y in terms of x. (3 marks) (DSE-2024 PI Q15)

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3. The coordinates of the points Q and R are $(10, -1)$ and $(-4, -9)$ respectively.

(a) Let P be a moving point in the rectangular coordinate plane such that $PQ = PR$. Denote the locus of P by Γ.

(i) Describe the geometric relationship between Γ and QR.

(ii) Find the equation of Γ. (3 marks)

(b) Let C be the circle which passes through Q, R and the point $(4, 3)$.

(i) Find the equation of C.

(ii) The coordinates of the point U are $(10, 4)$. It is found that U lies outside C. UV and UW are the tangents to C at the points V and W respectively. Is the area of the circumcircle of $\triangle UVW$ greater than 100? Explain your answer. (5 marks)

(8 marks) (DSE-2024 PI Q17)

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4. (a) $PQRS$ is a thin quadrilateral metal sheet, where $PQ = 12\,\text{cm}$, $PS = 10\,\text{cm}$, $QR = 13\,\text{cm}$, $\angle QPS = 82^\circ$ and $\angle QRS = 65^\circ$. Find

(i) the length of QS,

(ii) $\angle RQS$. (4 marks)

(b) The metal sheet $PQRS$ described in (a) is now folded along QS (see Figure 2). It is given that the angle between the plane PQS and the plane QRS is 80°.

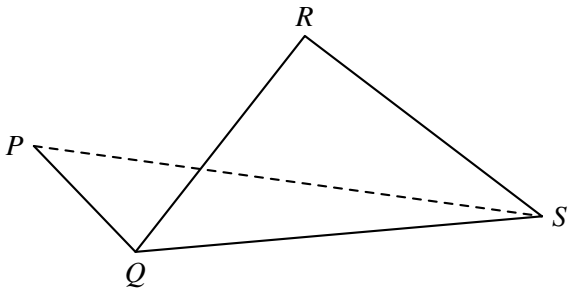


Figure 2

(i) Find the shortest distance from R to the plane PQS.

(ii) Let X be any point lying on the plane QRS. Someone claims that the distance between P and X exceeds $8\,\text{cm}$. Is the claim correct? Explain your answer. (4 marks)

(8 marks) (DSE-2024 PI Q18)

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5. Let $f(x) = 2x^2 + 4mx + 8x + 2m^2 + 8m + n$, where m and n are real constants such that $mn < 0$. Denote the vertex of the graph of $y = f(x)$ by P.

(a) Using the method of completing the square, express the coordinates of P in terms of m and n. (2 marks)

(b) Describe the geometric meaning represented by transforming $f(x)$ to $f\left(\frac{x}{5}\right) + 7$. (2 marks)

(c) Denote the vertex of the graph of $y = f\left(\frac{x}{5}\right) + 7$ by Q. Let (a_1, b_1) and (a_2, b_2) be the coordinates of P and Q respectively. It is given that a_1, $1 + n$, a_2 is an arithmetic sequence and b_1, $4 - m$, b_2 is a geometric sequence.

(i) Find the coordinates of P and Q.

(ii) The coordinates of the points R and S are $(3t + 27, t)$ and $(3t + 3, 2t - 3)$ respectively, where t is a real number. Is it possible that $PQRS$ is a rhombus? Explain your answer. (8 marks)

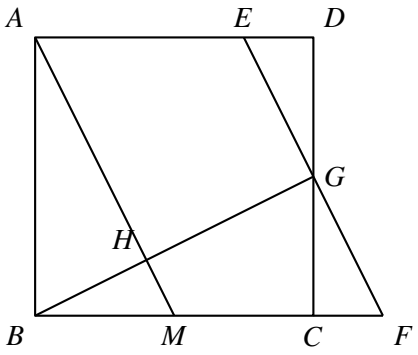
(12 marks) (DSE-2024 PI Q19)

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6. In the figure, $ABCD$ is a square. Let M be the mid-point of BC. E is a point lying on AD such that $AE : ED = 3 : 1$. F is a point lying on BC produced such that $EF//AM$. CD and EF intersect at the point G while AM and BG intersect at the point H. If the area of $\triangle BMH$ is $4\,\text{cm}^2$, then the area of the trapezium $AEGH$ is



A. $12\,\text{cm}^2$.

B. $33\,\text{cm}^2$.

C. $39\,\text{cm}^2$.

D. $45\,\text{cm}^2$.

(2 marks) (DSE-2024 PII Q17)

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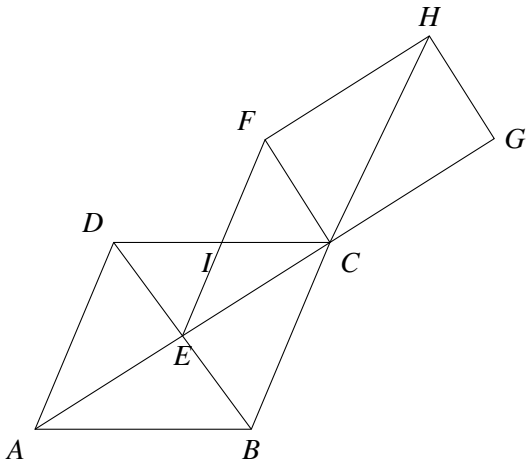
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7. In the figure, $ABCD$ is a rhombus. Denote the point of intersection of AC and BD by E. Let F be a point such that $BH//EF$ and $CFHG$ is a rectangle, where G and H are points lying on AC produced and BC produced respectively. Denote the point of intersection of CD and EF by I. Which of the following must be true?

I. $CI = FI$

II. $\angle ABE = \angle GCH$

III. $\triangle ADE \cong \triangle HCF$



A. I and II only

B. I and III only

C. II and III only

D. I, II and III

(2 marks) (DSE-2024 PII Q21)

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8. Denote the circle $3x^2 + 3y^2 - 6x + 12y - 4 = 0$ by C. Which of the following are true?

I. The origin lies inside C.

II. The circumference of C is less than 16.

III. The perpendicular distance from the centre of C to the x-axis is 2.

A. I and II only

B. I and III only

C. II and III only

D. I, II and III

(2 marks) (DSE-2024 PII Q27)

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9. $AF000000000BC_{16} =$

A. $175 \times 16^{11} + 188$.

B. $192 \times 16^{11} + 205$.

C. $175 \times 16^{12} + 188$.

D. $192 \times 16^{12} + 205$.

(2 marks) (DSE-2024 PII Q32)

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10. If $x = \log_2 y - 2$ and $(\log_2 y)^2 = 5\log_2 y + x - 7$, then $y =$

A. 1.

B. 8.

C. 1 or 3.

D. 3 or 8.

(2 marks) (DSE-2024 PII Q33)

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11. Consider the following system of inequalities:

$$\begin{cases} x-2y \leq 1 \\ x+4y \leq 13 \\ 2x-y \geq -1 \end{cases}$$

Let R be the region which represents the solution of the above system of inequalities. Find the constant c such that the least value of $5x-2y+c$ is 22, where (x,y) is a point lying in R.

A. 1

B. 23

C. 25

D. 29

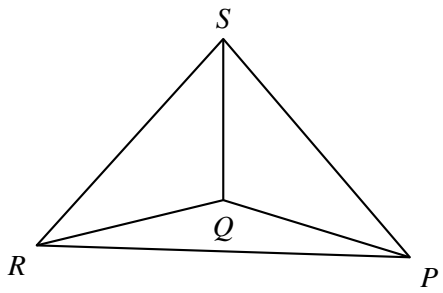
(2 marks) (DSE-2024 PII Q37)

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12. In the figure, the base PQR of the tetrahedron $PQRS$ lies on the horizontal ground. It is given that Q is vertically below S. If $\angle PQR = 90°$, $\angle QPS = 30°$ and $\angle QRS = 45°$, then $\cos\angle PRS =$



A. $\frac{1}{2}$.

B. $\frac{3}{4}$.

C. $\frac{\sqrt{2}}{4}$.

D. $\frac{\sqrt{3}}{6}$.

(2 marks) (DSE-2024 PII Q40)

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13. Let G, H, I and J be the centroid, the orthocentre, the in-centre and the circumcentre of $\triangle PQR$ respectively. If $\angle PQR = \angle PRQ = 22°$, which of the following are true?

I. G lies inside $\triangle PQR$.

II. H lies outside $\triangle PQR$.

III. I, J and Q are collinear.

A. I and II only

B. I and III only

C. II and III only

D. I, II and III

(2 marks) (DSE-2024 PII Q41)

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14. There are three questions in a test. The probabilities that a student answers the first question correctly, the second question correctly and the third question correctly are 0.6, 0.7 and 0.8 respectively. Find the probability that the student answers at least 1 question correctly in this test.

A. 0.024

B. 0.188

C. 0.812

D. 0.976

(2 marks) (DSE-2024 PII Q43)

I can finish _____ questions

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